7 (a) In 1913, Niels Bohr combined Planck's quantum theory, Einstein's concept of the photon, Rutherford's planetary model of the atom and Newtonian mechanics to derive a new model for the hydrogen atom. Several postulates were put forth.

Postulate 1. The electron moves in circular orbit, with radius *r*, around the proton due to the electric force of attraction, as shown in Fig. 7.1. In his model, the proton was considered to be stationary because its mass was very large compared to that of the electron.

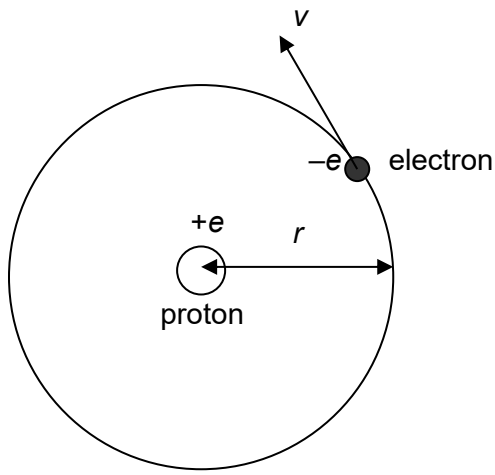

Fig. 7.1

Postulate 2. The angular momentum of the electron was quantised and this led to the hydrogen atom having allowable energies given by

$$E_n = -\frac{13.6}{n^2} \qquad \text{..... equation (1)}$$

where n = 1, 2, 3, …

Postulate 3. The atom emits radiation or photon when the electron made a transition from a higher energy to a lower-energy orbit. The frequency *f* of the photon emitted is related to the change in the energy of the atom rather than the orbital motion of the electron.

(i) With reference to postulate 3, write a word equation relating the change in the energy of the atom and the energy of the photon emitted.

..... [1]

(ii) Hence, with reference to postulate 3, explain the characteristic emission spectrum of hydrogen atoms.

.....

.....

.....[2]

[Turn over

(iii) By considering matter waves in the orbital motion of the electron, the radii of the orbits were found to be quantized. The smallest radius is 5.29×10^{-11} m.

1. By considering the forces acting on an electron, show that the momentum p of the electron in an orbit of radius r is given by

$$p = \sqrt{\frac{me^2}{4\pi\varepsilon_o r}}$$

where m is the mass of the electron.

[3]

2. Hence, determine the wavelength of the electron when $r = 5.29 \times 10^{-11}$ m.

wavelength = …….m [2]

(iv) Explain why this model of the hydrogen atom, where the electron orbits in a circular path with a fixed radius, will violate the Heisenberg uncertainty principle.

……

…….. [1]

(b) When high-energy electrons or any other charged particles bombard a metal target, x-rays are emitted.

The origin of the characteristic x-rays is due to the removal of the inner-shell electron from the atom by the bombarding electrons that have sufficient energies. The vacancy that is created in the inner shell is filled when an electron that is at a higher energy level in the atom transits into the energy level containing the vacancy. Fig. 7.2 shows the shells (*K*, *L*, *M*, *N* and *O*) present in the atom and how the characteristic x-rays, K-lines and other peaks, are obtained.

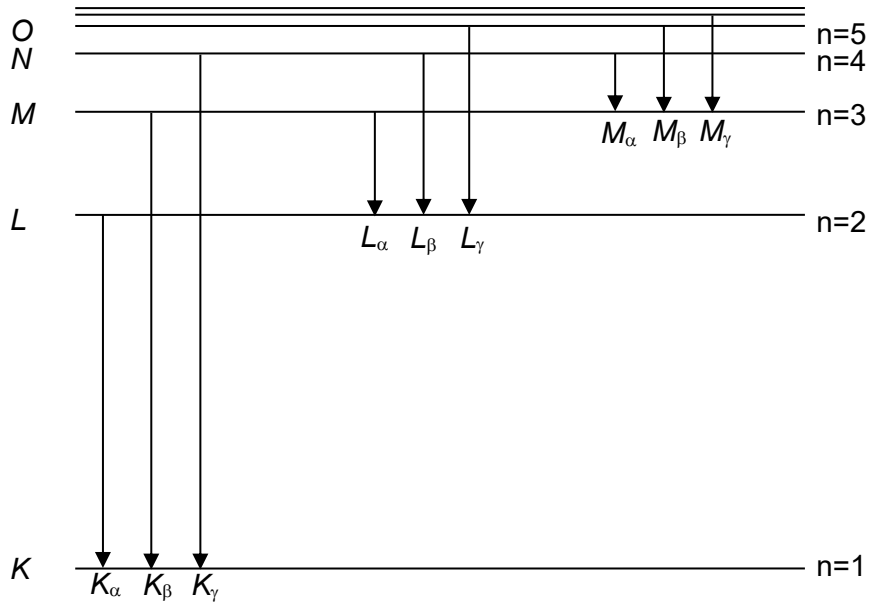

Fig. 7.2

Equation (1) in part **(a)** may be modified to describe the energy for the *L*-shell. Due to the presence of one *K*-shell electron, the other electron in *L*-shell will 'see' an effective nuclear charge of approximately $(Z - 1)e$, where e is the elementary charge and Z is the atomic number of the element.

As such equation (1) in part **(a)** can be approximated as

$$E_n = -\frac{(13.6\ eV)(Z-1)^2}{n^2}$$

(i) For an electron that makes a transition from the *L* shell (with n = 2) to the *K* shell (with n = 1), derive the relationship between the frequency f of the radiation and Z.

[3]

[Turn over

(ii) It is thought that the wavelength λ of K_α x-ray radiation, varies with atomic number of element *Z* according to the expression below as suggested by Mosley in 1914.

$$\sqrt{\frac{1}{\lambda}} = A(Z - B) \text{ where } A \text{ and } B \text{ are constants}$$

The frequency *f* of K_α x-ray radiation, of a few elements are given in Fig. 7.3.

Element	Atomic number *Z*	Frequency *f* / 10^{18} Hz	$\sqrt{\frac{1}{\lambda}}$ / 10^4 m$^{-½}$
Titanium	22	1.08	6.01
Chromium	24	1.30	6.59
Iron	26	1.54	7.16
Nickel	28	1.79	
Zinc	30	2.07	8.30
Gallium	31	2.21	8.59

Fig. 7.3

Complete Fig. 7.3 for Nickel. [1]

(iii) The data from Fig. 7.3 are used to plot the graph of Fig. 7.4.

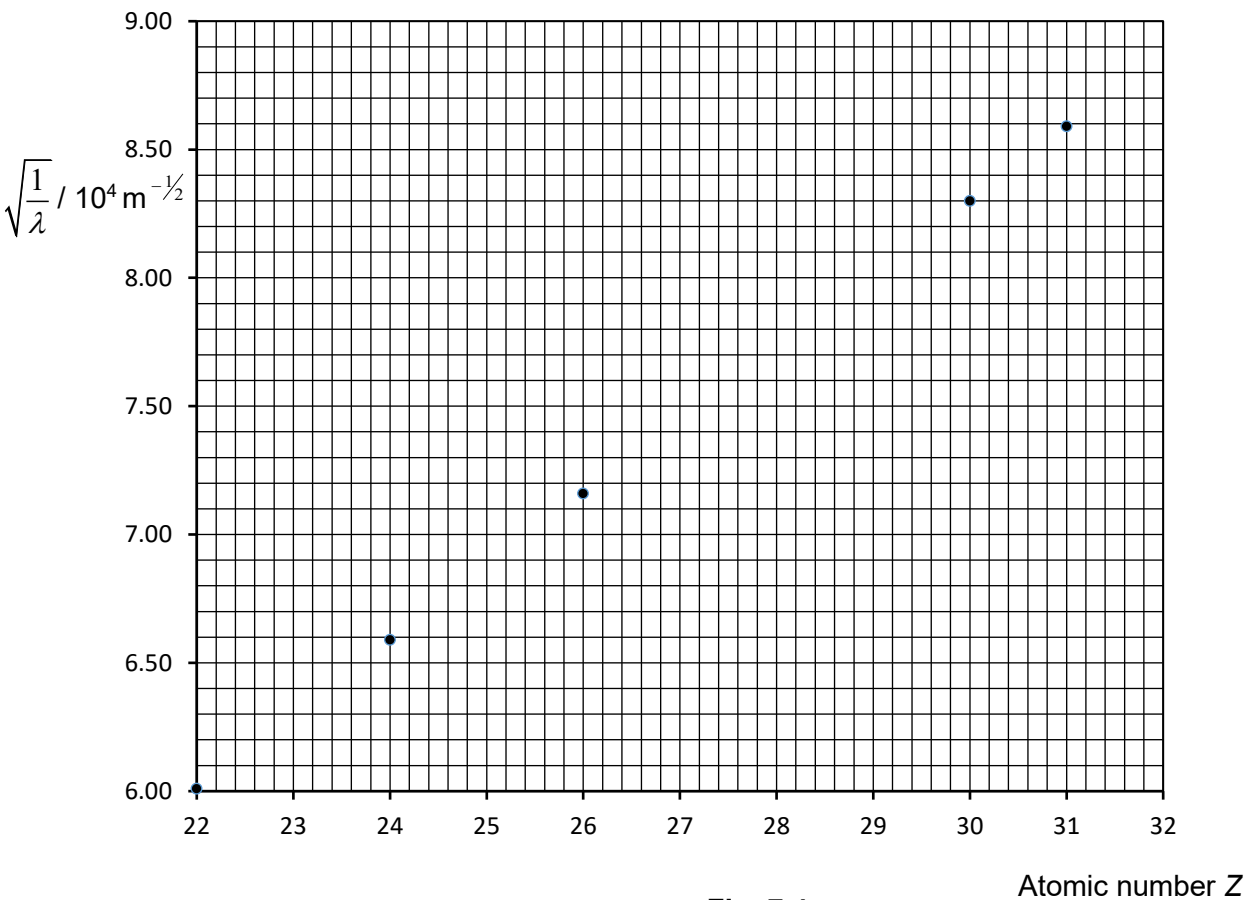

Fig. 7.4

On Fig. 7.4,

1 Plot the point corresponding to Nickel. [1]

2 Draw the best-fit line for all the points. [1]

[Turn over

(iii) Use the line drawn in **Fig. 7.4** to determine the magnitude of the constants *A* and *B* in the expression in **b(ii)**.

A = ……..$m^{-1/2}$

B = ……[3]

(c) One of the possible use for characteristic spectrum is to allow impurity in specimens to be detected by scientists. A cobalt target is bombarded with electrons, and the wavelengths of its characteristic x-ray spectrum are measured. There is also a second fainter characteristic spectrum, which is due to an impurity in the cobalt. The wavelengths of the K_α lines are 178.9 pm (cobalt) and 143.5 pm (impurity), and the atomic number *Z* for cobalt is 27.
Using Fig. 7.3, deduce the impurity in the cobalt.

[2]